Depth dependence, internal robustness, and Bayesian neural-network learning

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Abstract

We investigate whether the exponential depth dependence in the existing width-independent Bayesian learning bounds is necessary, and which natural robustness restrictions remove it. The known norm-capacity obstruction does not establish an exponential sample lower bound for a small target under the specified Gaussian prior; constants and single-coordinate targets in fact have depth-independent sample bounds in the strong-prior limit. We strengthen the capacity obstruction: even global polynomial input-Lipschitz bounds allow all Boolean truth tables at polynomial weighted norm and linear depth. Internal robustness is different. An RMS-normalized all-layer margin r, together with energy K, permits compression to width $O(Kr^{-2}(d + \log L))$. A second argument tracks entropy only in unprocessed original weights while compressing a posterior. It yields one polynomial-depth sample bound, under a single norm-calibrated robustness restriction, simultaneously for every $\gamma \geq 1$ and every width above the width of a layered Boolean circuit. Neither a hard energy cutoff nor a width-first limit is required. An alternative calibration uses the guaranteed noise radius of a nonexpansive network with the same depth and readout. Established all-layer-margin and noise-stability literature supports the naturalness of the condition, but not its universal validity for ordinary gradient-trained networks.

Status and scope. This is a research report with mathematical derivations and independent proof reviews by other AI agents, not a Lean-certified extension of the published main results. We retain the original all-input clean margin and posterior-confidence conventions unless explicitly stated. The attached Claude note [1] is the starting point for robust compression; the Bernstein improvement, statistical bridges, and counterexample refinements are worked out here. The explicit bounds are deliberately loose and not proposed as practical numerical estimates. No claim of priority over the broader robustness/compression literature is made.

Assessment of the attached note. The exact-simulation argument and its main robust-subsampling proof check out; we found no major error in that proof. Its necessity discussion concerns uniform norm compression and counting, rather than a proved sample lower bound for the specified Gaussian posterior. Its robust/plain complexity sandwich also does not identify arbitrary analog complexity with robust complexity at the same architecture. The present report supplies explicit robust circuit witnesses and the additional Gaussian posterior arguments.

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1 Findings and unresolved questions

There are three distinct questions.

1. *Can the old uniform norm-counting argument have polynomial depth?* No. At depth $L = d + O(1)$ and width 2^{d+1}, polynomial weighted norm already contains every Boolean truth table. This remains true with a global input Lipschitz constant polynomial in d.

2. *Does the exact unrestricted Gaussian Bayesian learner therefore need exponentially many samples for a polynomial-complexity target?* This does not follow, and remains unresolved in this investigation. The missing step is a target-specific prior/posterior mass or minimum-energy comparison.

3. *Can a natural internal robustness condition give polynomial depth?* Yes, under precise support and target hypotheses. An all-layer margin is an established notion rather than a property invented solely for this proof. Saturated Boolean circuits satisfy an inverse-polynomial version without depth-dependent error accumulation.

The resulting learning statements are summarized below. In the unified theorem, "polynomial" means one polynomial whose degree and coefficients are independent of both depth and $\gamma \geq 1$.

Regime	Additional support condition	Result established here
Every $\gamma \geq 1$, including mean field	Fixed all-layer margin, or either calibrated support rule in Sections 4.3 and 4.6	One polynomial bound; every width above circuit width
Strong-prior limit	The same margin conditions	Polynomial depth; in fact every Gaussian scale $\sigma^2 \leq 1/N$
Any variance, including $\gamma = 1$	All-layer margin and a width-independent hard energy budget	Polynomial depth by a finite-class bound
Unrestricted Gaussian prior	Original global clean margin only	General necessity unresolved; constants and dictators have a depth-independent strong-limit bound

The circuit targets in this table are ordinary bounded-fan-in Boolean circuits, expressed in layers. They are not silently identified with arbitrary analog tanh networks at exactly the same depth. Polynomial-overhead digital simulation gives an expressivity statement, but can increase both depth and minimum required width; see Appendix C.

2 Model and an all-layer margin

Let $L \geq 2$, $d \geq 1$, and let every hidden layer have width N. The input is $h^0 = (1, x)$ for $x \in \{-1, 1\}^d$; there are no subsequent biases. Set

$$h^s = \phi(W^s h^{s-1}) \quad (1 \leq s < L), \qquad z_w(x) = v h^{L-1},$$

where $|\phi| \le 1$ and ϕ is 1-Lipschitz. Circuit constructions and finite-variance target-mass estimates use $\phi = \tanh$. The weighted energy is

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_F^2 + N\|v\|_2^2.$$

For auxiliary classifiers produced by rounding, set $\operatorname{sign}(0) = +1$. The strong-prior Gaussian density in weighted coordinates is proportional to $\exp(-\kappa_N/(2\sigma^2))$. At exponent γ, the independent entry variances are $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-1-\gamma}$ for the first, internal, and readout layers respectively.

Use additive *postactivation* perturbations:

$$h_\eta^s = \phi(W^s h_\eta^{s-1}) + \eta^s, \qquad z_w(x;\eta) = v h_\eta^{L-1} + \eta^L.$$

Their total normalized size is

$$\|\eta\|_*^2 = \sum_{s<L}\frac{\|\eta^s\|_2^2}{N} + |\eta^L|^2. \tag{1}$$

The factor N^{-1} measures hidden noise by its RMS amplitude and makes the condition compatible with neuron cloning.

Definition 2.1. *A network has all-layer sign margin r for f if*

$$f(x)z_w(x;\eta) > 0 \quad \text{for every } x \in \{-1,1\}^d \text{ and every } \|\eta\|_* \le r.$$

Let $\mathcal{R}_{r,N,f}$ additionally require the original clean margin $f(x)z_w(x) > 1/2$ for every x. The restricted prior is the raw Gaussian conditioned on $\bigcup_f \mathcal{R}_{r,N,f}$, followed by conditioning on the observed noiseless labels.

The condition refers to a network's own clean signs; it does not reveal unknown target labels to the prior. Its simultaneous finite-radius character is important. A derivative evaluated only on the clean computation, random noise tolerance, or robustness at one layer at a time is weaker.

This is an absolute-RMS version of the all-layer-margin idea of Wei and Ma [3]. Their normalization uses hidden-state norms, rather than the fixed $\sqrt{N}$ used here. The two are not automatically equivalent when hidden activity can collapse. The attached note instead gives each layer its own RMS budget τ and retains output margin $1/3$. The total norm and maximum per-layer norm differ by at most $\sqrt{L}$; either convention can give polynomial depth. The sharper proof below allocates noise by layer energy and avoids even that extra factor in compressed width.

For a posterior Π, write

$$R_D(\Pi) = \mathbb{E}_{x\sim D}\mathbb{P}_{g\sim\Pi}\{g(x) \ne f(x)\}.$$

The fraction of test inputs with correct-label probability at most $2/3$ is at most $3R_D(\Pi)$. All learning bounds below therefore imply the original confidence guarantee by replacing the desired error parameter by one third of its value. They are not new majority-vote lower bounds.

3 Compression with polynomial depth dependence

Theorem 3.1. *Suppose $\kappa_N(w) \le K$, $K \ge 1$, and w has all-layer sign margin $0 < r \le 1$ for f. Put*

$$m = \left\lceil 256Kr^{-2}[d + \log(400L)] \right\rceil.$$

Then f has a width-m, depth-L implementation with clean margin $> 3/4$, energy at most $4K/r^2$, and every weight bounded in absolute value by m. The compressed network need not itself retain the original robustness radius.

Proof. Assume first that $N \geq m$. For $2 \leq s < L$ put $k_s = \|W^s\|_F^2$, and put $k_L = N\|v\|_2^2$, so $\sum_{s=2}^{L} k_s \leq K$. Zero-energy layers need no approximation. In each internal row, and in the readout row, delete entries exceeding

$$\frac{8\sqrt{K}\,\|\text{row}\|_2}{r\sqrt{N}}$$

in absolute value. Leave the first layer unchanged. Denote the pruned weights by bars. Against any vector with coordinates in $[-1,1]$, the internal pruning error has norm at most $r\sqrt{Nk_s/K}/8$; the readout error is at most $r\sqrt{k_L/K}/8$. Indeed, the removed row ℓ^1 mass is at most its squared ℓ^2 norm divided by the deletion threshold.

Independently sample m neurons with replacement from each hidden layer. Keep sampled first-layer rows, and multiply all subsequent sampled source weights by N/m. Define a virtual full-width forward pass that uses the sampled sources at each layer. Its sampled coordinates are exactly the small network's coordinates. Its layer-$s-1$ activation depends only on samples through layer $s-2$, and is independent of the fresh source sample used for layer s. This conditional independence is the central point in the attached note's exact-simulation argument.

Fix an input and condition on those earlier samples. A fresh contribution to an internal preactivation vector is $Y = N\bar{W}^s_{\cdot J}\widetilde{h}^{s-1}_J$, with J uniform on $[N]$. Pruning and bounded activations give

$$\mathbb{E}\|Y - \mathbb{E}Y\|_2^2 \leq Nk_s, \qquad \|Y - \mathbb{E}Y\|_2 \leq \frac{16\sqrt{NKk_s}}{r}.$$

For centered independent Hilbert-space variables of norm at most B and total second moment at most V, Bernstein's inequality gives

$$\mathbb{P}\left\{\left\|\sum_i Y_i\right\| \geq t\right\} \leq 2\exp\left(-\frac{t^2}{2(V + Bt/3)}\right).$$

The dimension-free form used here is recorded, for example, in Tarrès–Yao [7], Appendix A, Corollary A.2, equation (A-1). Applying it at sample-mean error $r\sqrt{Nk_s/K}/8$ gives failure probability at most

$$2\exp\left(-\frac{3mr^2}{640K}\right) \leq 2\exp\left(-\frac{mr^2}{256K}\right).$$

The readout calculation is identical with the factors $\sqrt{N}$ removed. A union bound over all 2^d inputs and at most L sampled layers costs at most $1/200$ for the stated m.

On success, pruning plus sampling contributes normalized error at most $r\sqrt{k_s/K}/4$ at layer s. Lipschitzness turns each preactivation error into an additive postactivation error without increase. Thus the virtual pass is an exact noisy pass of the original network with

$$\|\eta\|_*^2 \leq \sum_{s=2}^{L} \frac{r^2 k_s}{16K} \leq \frac{r^2}{16}.$$

The expected weighted norm of the sampled network is at most K: sampling is unbiased for every squared-norm term, with the prescribed width rescalings. Markov's inequality gives norm at most $4K$ with probability at least $3/4$. The good-noise and good-norm events consequently intersect with probability at least $3/4 - 1/200$.

Every realized signed output in that intersection is $> 3r/4$. Otherwise, an additional scalar output perturbation of size at most $3r/4$ would bring it to zero, with total noise norm at most r, contradicting the all-layer sign margin. Scale the readout by $1/r$ to obtain margin $> 3/4$ and energy at most $4K/r^2$.

If $N < m$, clone the original network to width m. Balanced multiplicities increase weighted norm by at most two and preserve the output. Its clean margin is $> r$, because output-only noise of size r is allowed. Scaling by $1/r$ gives the desired conclusion. Finally, $m \geq 4K/r^2$ makes the energy bound imply that each individual weight has magnitude at most m. □

Define $P = m(d+2) + (L-2)m^2$ and the conservative entropy bound

$$H(d, L, K, r) = P \log[3 + 8m(d + L + 1)(2m)^{2L}], \tag{2}$$

with m from Theorem 3.1. A parameter grid of sufficiently small mesh preserves the $> 3/4$ margin: a safe output perturbation bound is $\delta(d+L+1)(2m)^{2L}$ when each parameter is changed by at most δ. The number of functions under the theorem's hypotheses is therefore at most $\exp H$. Crucially, an exponential numerical sensitivity appears only *inside a logarithm*. In particular,

$$H = O\left(L^2 K^2 r^{-4} (d + \log L)^2 \log \frac{K(d + \log L)}{r}\right)$$

up to universal changes of constants and harmless additions inside logarithms.

4 Learning and nonvacuous circuit targets

4.1 Robust digital gates

Consider a layered Boolean circuit with at most C wires per stage, including a constant $+1$ rail. Carry unused wires by identity gates and fill unused positions by constants. AND, OR, NOT and COPY are sign functions of $a + b - c$, $a + b + c$, $-a$, and a, where $c = +1$. Replace each linear form by its tanh after multiplication by 8, and use readout coefficient 2 on the answer wire.

If every incoming noisy coordinate differs from its Boolean value by at most $1/8$, each intended signed preactivation is at least 5. The tanh output then has error $< 1/16$. Adding a further coordinate perturbation of magnitude at most $1/16$ restores the same invariant. Thus errors do not accumulate with depth. A base RMS radius $1/(16\sqrt{C})$ suffices, its energy is at most

$$B_0 = 576d + 576(L-2)C + 4C = O(d + LC),$$

and its noisy output margin exceeds one. Repeated inputs can give an edge of magnitude 16; pad to $C \geq 16$ if the bounded-weight convention requires weights no larger than width.

At every $N \geq C$, clone the wires equally and leave dead padding. Clone-group means follow the original computation with averaged injections. The normalized total noise increases by at most $\sqrt{2}$ under this map, and the norm increases by at most two. Hence every such learner width has a target witness of energy at most $2B_0$ and all-layer radius at least $1/(16\sqrt{2C})$. This is an explicit robust target construction, rather than an assumption that ordinary analog complexity automatically equals robust complexity.

4.2 Strong-prior theorem

For fixed r, let $b_{r,N}(g)$ be the infimum of κ_N on $\mathcal{R}_{r,N,g}$. These implementation events are open: the noise ball is compact, the input set is finite, and the minimum noisy signed output is continuous in the weights. The usual Gaussian small-variance argument therefore gives

$$\lim_{\sigma \downarrow 0} 2\sigma^2 \log \mu_{N,\sigma}(\mathcal{R}_{r,N,g}) = -b_{r,N}(g).$$

The normalization of the robustness-conditioned prior cancels in ratios. For completeness, this rate calculation is elementary in the fixed finite-dimensional parameter space. After a linear change of coordinates, κ_N is a squared Euclidean norm. The Gaussian Chernoff bound with parameter $(1-\xi)/(2\sigma^2)$ gives rate at most $-(1-\xi)\inf_U \kappa_N$ for any measurable event U; let $\xi \downarrow 0$. For an open nonempty U, a small ball about a point of energy within η of its infimum has positive volume and maximum energy arbitrarily close to that infimum. Integrating the Gaussian density on that ball gives the matching lower rate as $\eta \downarrow 0$.

Theorem 4.1. *Let $0 < r \leq 1$, $K \geq 1$, and $\varepsilon, \delta \in (0,1)$. Suppose $b_{r,N}(f) \leq K$. There is a threshold $\sigma_0 > 0$, depending on $d, L, N, r, \phi, f, \varepsilon$ but not on the input distribution, sample size, or δ, such that for every $0 < \sigma < \sigma_0$ it suffices to take*

$$n \geq \frac{2}{\varepsilon}\left[H(d, L, K+1, r) + \log\frac{1}{\delta}\right]$$

to obtain posterior average error at most ε, with probability at least $1-\delta$ over IID training examples from any input distribution. The sufficiently-small-variance threshold may depend on width.

Proof. There are finitely many Boolean functions. The total prior mass of functions with $b_{r,N}(g) > b_{r,N}(f)$, divided by the target mass, tends to zero. The functions with rate no larger than the target have log cardinality at most $H(d, L, K+1, r)$, by near-minimizers and Theorem 3.1. A union bound excludes every member of this finite class with error $> \varepsilon/2$ at the stated sample size. The remaining posterior mass outside the class is at most $\varepsilon/2$, since the target is always consistent. This proves the claim. □

The digital construction makes this bound polynomial jointly in d, L, C at inverse-polynomial r, uniformly over $N \geq C$ in the original order of limits. A fixed radius prior covers a specified class of circuit widths; it is not silently changed for each unknown target.

4.3 One fixed prior without a target-size choice

A possible calibrated support condition uses an aggregate of layer norms, rather than their product. Define

$$S_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_{\mathrm{op}}^2 + N\|v\|_2^2 \leq \kappa_N(w).$$

Require clean margin $> 1/2$ together with all-layer sign robustness at radius

$$r(w) = \frac{c}{\sqrt{1+S_N(w)}}, \qquad c = 1/100.$$

This is a proposed prior design, rather than an empirically established normalization. It expresses an inverse-polynomial internal precision requirement relative to an aggregate measure of layer amplification. The rule is independent of the target, its size, the data, and γ. It charges for internal precision rather than forbidding particular Boolean functions outright: increasing the readout norm can lower the required radius, but the Gaussian prior charges for that increase. It remains open in parameter space: write the perturbation as $r(w)u$ over the fixed compact unit ball, and use continuity of S_N. On the energy sublevel $\kappa_N \leq B+1$, every member has radius at least $c/\sqrt{B+2}$, so the compression and counting bounds are polynomial in d, L, B. The cloned gate witness has $S_N \geq 4C$ from its readout alone, making the required radius smaller than its certified radius. Its energy is at most $2B_0$. Thus this *single* support rule gives the strong-prior polynomial-depth

conclusion for all layered Boolean circuits, without a target-complexity-dependent choice of r. The same rule also works uniformly at every $\gamma \geq 1$, as proved below. Using κ_N instead of S_N in this calibration would suffice in the strong-prior limit, but would not give the same finite-variance argument throughout $1 < \gamma < 2$.

4.4 Every fixed exponent $\gamma > 1$

Theorem 4.2. *Fix $\gamma > 1$. Let f be computed by a layered fan-in-two Boolean circuit of width C and $L-1$ stages, and let $0 < r \leq 1/(64\sqrt{2C})$. Condition the specified Gaussian prior on $\bigcup_g \mathcal{R}_{r,N,g}$. For every learner width $N \geq C$, polynomially many samples in*

$$d,\ L,\ C,\ 1/r,\ 1/\varepsilon,\ \log(1/\delta)$$

suffice for posterior average error at most ε with probability $1-\delta$. The polynomial degree may depend on γ, but not on L.

Here are the probability and robustness steps; explicit constants for the target-mass bound appear in Appendix A. The robust digital cylinder gives, for every $\gamma \geq 1$,

$$\mu_{N,\gamma}(\mathcal{R}_{r,N,f}) \geq e^{-DN^\gamma}, \qquad D = \mathrm{poly}(d, C, L). \tag{3}$$

For a tail parameter $K \geq 1$, put $a = r/(4\sqrt{L})$. The Gaussian singular-value-tail argument in the existing $\gamma > 1$ paper implies that, once

$$N^{\gamma-1} \geq \max\{8(d+1), 64/a^2\},$$

discarding singular values at most a from the internal matrices leaves energy at most LK, outside raw probability $Le^{-KN^\gamma/8}$.

Unlike the old proof, no downstream product of operator norms is needed. Along any radius-$r/2$ noisy pass of the truncated network, hidden states have norm at most $(1+r/2)\sqrt{N}$. Each discarded matrix produces an additional normalized postactivation error at most $a(1+r/2)$. Combining all layers gives total noise at most

$$r/2 + \sqrt{L}\, a(1+r/2) \leq 7r/8 < r.$$

That exact trajectory is therefore an allowed noisy pass of the original draw. The truncated network retains all-layer sign margin $r/2$, though not necessarily clean margin $1/2$. Theorem 3.1 restores the clean margin and places its sign function in a class with entropy $H(d, L, LK, r/2)$.

Choose

$$K = 8[D + \log(2L/\varepsilon)], \qquad T = \left\lceil \max\{C, [8(d+1)]^{1/(\gamma-1)}, (32\sqrt{L}/r)^{2/(\gamma-1)}\} \right\rceil.$$

For $N \geq T$, the exceptional mass divided by target mass is at most $\varepsilon/2$. For $C \leq N < T$, the target-mass-only posterior bound costs at most DT^γ. Consequently the sufficient bound

$$n \geq \frac{2}{\varepsilon}\left[\max\{H(d, L, LK, r/2), DT^\gamma\} + \log\frac{2}{\varepsilon\delta}\right]$$

proves the theorem. Every displayed term has a polynomial majorant of the claimed form. These estimates deteriorate as $\gamma \downarrow 1$; Section 4.5 gives a different, uniform argument.

Corollary 4.3. *The single calibrated support condition of Section 4.3 has polynomial sample complexity jointly in d, L, C at every fixed $\gamma > 1$, for every learner width $N \geq C$, for the same circuit targets. There is no target-size-dependent robustness parameter or hard norm cutoff.*

Proof. The target cylinder from Appendix A still lies in the calibrated support. Indeed its readout center satisfies $\sqrt{N}\|v_*\| \geq \sqrt{2C}$ and the cylinder changes this norm by at most $1/4$. Therefore $S_N \geq (\sqrt{2C} - 1/4)^2 \geq C$ throughout the cylinder, and $r(w) \leq c/\sqrt{1+C} < 1/(64\sqrt{2C})$. Its raw mass is still at least e^{-DN^γ}.

Choose $K = 8[D + \log(2L/\varepsilon)]$, put $r_0 = c/\sqrt{1+LK}$, and use cutoff $a = r_0/(4\sqrt{L})$. On the same Gaussian good event as in Theorem 4.2, the first and readout energy terms are at most K, and each internal matrix's retained singular values have squared sum at most K. Every original singular value is then at most $\max\{\sqrt{K}, a\} = \sqrt{K}$, so $S_N \leq LK$. Thus every supported good draw has robustness radius at least r_0, and its truncation has energy at most LK and radius $r_0/2$. Its own calibrated radius need not be retained; the fixed-radius compression theorem is all that is required.

Use the preceding sample bound with $H = H(d, L, LK, r_0/2)$ and

$$T = \left\lceil \max\left\{ C, [8(d+1)]^{1/(\gamma-1)}, \left[\frac{1024L(1+LK)}{c^2}\right]^{1/(\gamma-1)} \right\} \right\rceil.$$

The target-relative tail and the small-width argument are unchanged. Every term has the claimed polynomial majorant. □

4.5 One theorem for all $\gamma \geq 1$

The preceding spectral truncation proof does not extend uniformly to mean field: the Gaussian internal bulk has operator norm of order one, and keeping all singular values above a small cutoff can leave energy of order N. Appendix F resolves this by compressing a posterior distribution, while tracking conditional entropy only in the original weights that have not yet been processed. All approximation errors become simultaneous activation perturbations of the original network. The proof allows arbitrary mixtures of functions in the posterior.

Theorem 4.4 (A single robustness restriction for all widths and scales)**.** *Use* tanh *activation and the support restriction of Section 4.3: global clean margin $1/2$ and all-layer sign robustness at radius $r(w) = 1/[100\sqrt{1+S_N(w)}]$. Let f be computed by a layered fan-in-two Boolean circuit over AND, OR, NOT, COPY, and constant gates, of width $C \geq 1$ with $L-1$ stages, including the constant rail. There is a fixed polynomial P such that, for any input distribution on $\{-1,1\}^d$, any $0 < \varepsilon, \delta < 1$, and*

$$n \geq P(d, L, C, \varepsilon^{-1}, \log(1/\delta)),$$

with probability at least $1-\delta$ over the noiseless sample, the posterior Gibbs error is at most ε, simultaneously for every $N \geq C$ and every raw Gaussian with entry variances $t^2/d, t^2/N, t^2/N^2$, $0 < t \leq 1$. The same polynomial, after a constant rescaling of ε, bounds the fraction of inputs where the posterior probability of the correct label is at most $2/3$, or at most $1/2$. In particular the result is uniform in all $\gamma \geq 1$, by taking $t^2 = N^{1-\gamma}$, and in every strong-prior scale $\sigma^2 \leq 1/N$. Neither coefficients nor degree of P depend on N, t, or γ.

Proof. Write $\mu = \mu_{N,t}$ for the raw Gaussian. The robust target cylinder of Appendix A has mass at least $\exp(-DN/t^2)$ for a $D = \mathrm{poly}(d, L, C)$, uniformly for $N \geq C$, $0 < t \leq 1$. Its radius exceeds the required calibrated radius, as checked in Section 4.3. For every sample s, the posterior Q_s is μ conditioned on the support restriction and consistency with s. Hence

$$D_{\mathrm{KL}}(Q_s \| \mu) \leq DN/t^2.$$

Set $\beta = \varepsilon/4$, $\alpha = \varepsilon/12$, and

$$B = \max\{16, 8(d+1), 8[D + \log(L/\beta)]\}, \quad G = \{S_N \le LB\}, \quad r_0 = \frac{1}{100\sqrt{1+LB}}.$$

The first-layer term of S_N is $(t^2/N)\chi^2_{N(d+1)}$ and the readout term is $(t^2/N)\chi^2_N$. Their Gaussian moment bounds, and $\mathbb{P}(\|W^s\|_{\mathrm{op}} > 2t + u) \le \exp[-Nu^2/(2t^2)]$ internally, give

$$\mu(G^c) \le L \exp[-NB/(8t^2)], \qquad Q_s(G^c) \le \beta.$$

Consequently $Q_s^G = Q_s(\cdot \mid G)$ has all-layer radius at least r_0, is still consistent, and has KL at most $N(D+1)/t^2$.

Theorem F.1 supplies one finite class $\mathcal{H}(d, L, D+1, r_0, \alpha)$, independent of N, t, s, with polynomial log-size H_0, and a law Γ_s on it. At each input, Γ_s and Q_s^G admit a prediction coupling with mismatch probability at most α. Thus the expected empirical error of Γ_s is at most α, and its true error differs from that of Q_s^G by at most α. With probability $1-\delta$, simultaneously for every $h \in \mathcal{H}$,

$$\mathrm{err}(h) \le 2\,\mathrm{err}_s(h) + \frac{8[H_0 + \log(1/\delta)]}{n}.$$

Indeed, violation implies empirical error below half the true error and true error above $8[H_0 + \log(1/\delta)]/n$; the multiplicative Chernoff bound and a union bound apply. Averaging under Γ_s gives

$$\mathrm{GibbsErr}(Q_s) \le \beta + 3\alpha + \frac{8[H_0 + \log(1/\delta)]}{n} \le \varepsilon$$

when $n \ge 16[H_0 + \log(1/\delta)]/\varepsilon$. All parameters of H_0 have polynomial bounds in the stated quantities. The same finite-class event works for every N, t. Finally, if $p_s(x)$ denotes posterior probability of the correct label, then $\mathbb{P}_x\{p_s(x) \le q\} \le \mathrm{GibbsErr}(Q_s)/(1-q)$. □

The same compression theorem gives a polynomial bound for a fixed all-layer radius r whenever the raw robust target mass is at least $\exp(-NK/t^2)$ with K polynomial in target parameters. This is recorded in Corollary F.4.

A simpler alternative with a hard energy cutoff. One may instead restrict the support further to $\kappa_N < K$, with K independent of learner width. Its entire function class has entropy at most $H(d, L, K, r)$. If the target has a strict robust witness of energy below K, it has positive prior mass at every nonzero variance. For

$$n \ge [H(d, L, K, r) + \log(1/\delta)]/\varepsilon,$$

every consistent supported function has error at most ε with probability $1-\delta$. This holds for any Gaussian variance, including mean field. This is a stronger change of prior, and is unnecessary in Theorem 4.4. In particular a raw energy budget that itself grows with width would not provide the advertised width independence.

4.6 A normalization with a nonexpansive-network interpretation

An alternative support rule has a simpler interpretation. Write $J = L - 1$, $E_v = N\|v\|_2^2$, and require the same global clean margin and all-layer sign robustness, now at radius

$$r_{\mathrm{ref}}(w) = \frac{1}{100\sqrt{1 + JE_v(w)}}.$$

The denominator is the total-noise gain bound for a network with nonexpansive hidden transitions and the same readout. This is a fixed fraction of its guaranteed noise radius, with no target-size or Gaussian-scale parameter and no hard energy cutoff.

Proposition 4.5. *If every internal hidden matrix satisfies* $\|W^s\|_{\mathrm{op}} \le 1$ *and the clean output margin exceeds* $1/2$*, this support condition holds. The first matrix need not be nonexpansive.*

Proof. Let a_s be the Euclidean hidden displacement. There is no input noise, so $a_1 \le \|\eta^1\|_2$; subsequently, $a_s \le a_{s-1} + \|\eta^s\|_2$ by nonexpansiveness. Thus

$$|z(x;\eta) - z(x)| \le \sqrt{E_v} \sum_{s=1}^{J} \frac{\|\eta^s\|_2}{\sqrt{N}} + |\eta^L| \le \sqrt{1 + JE_v}\, \|\eta\|_*.$$

At the specified radius this is at most $.01 < 1/2$. □

The support itself permits larger operator norms whenever the actual finite-radius robustness suffices. Hence this is an interpretation and a sufficient architectural condition, not an additional spectral restriction.

Corollary 4.6. *All conclusions of Theorem 4.4 also hold for this readout-reference support: one polynomial sample bound simultaneously for every* $N \ge C$ *and every* $0 < t \le 1$*, including all* $\gamma \ge 1$*.*

Proof. Every draw in the target cylinder of Appendix A has $E_v \ge C$ and sign robustness at radius $\tau_0 = 1/(64\sqrt{2C})$. Since $J \ge 1$,

$$r_{\mathrm{ref}}(w) \le .01/\sqrt{1 + C} < \tau_0.$$

The same target mass lower bound $\exp(-DN/t^2)$ therefore applies. For $0 < \beta \le 1/2$, put

$$B = \max\{8, 8[D + \log(1/\beta)]\}, \qquad G = \{E_v \le B\}.$$

Under the raw Gaussian, $E_v = (t^2/N)\chi^2_N$. Its exponential moment gives

$$\mu_{N,t}(G^c) \le \exp[-NB/(8t^2)], \qquad Q_s(G^c) \le \beta,$$

where the second bound divides by the target cylinder mass. The consistent conditional law $Q_s(\cdot \mid G)$ has radius at least $r_0 = .01/\sqrt{1 + JB}$ and KL at most $N(D+1)/t^2$. Theorem F.1 and the same finite-class argument as in Theorem 4.4 give

$$\mathrm{GibbsErr}(Q_s) \le \beta + 3\alpha + 8[\log|\mathcal{H}| + \log(1/\delta)]/n,$$

with one class $\mathcal{H}(d, L, D+1, r_0, \alpha)$ independent of N, t. The earlier choices of α, β, n prove the claim. □

This normalization is less tied to the intermediate spectral estimates than S_N. The two required radii have no universal ordering: that ordering depends on whether S_N is larger or smaller than JE_v. Both are proposed prior designs, not established laws of ordinary training. In our saved toy models, the readout-reference radius is always smaller than the spectral-calibrated radius, so every successful certificate reported below transfers to it.

5 Naturalness and what trained networks are known to satisfy

There is a direct numerical interpretation. Rounding every hidden activation and the scalar output with absolute error at most 2^{-b} creates total normalized noise at most $\sqrt{L}\,2^{-b}$. An all-layer margin r therefore preserves every label whenever

$$b \geq \left\lceil \log_2(\sqrt{L}/r) \right\rceil .$$

An inverse-polynomial margin means that only logarithmically many fractional bits of activation precision are needed. This is a consequence of robust computation; choosing a quantized activation rule alone does not establish the adversarial finite-radius property.

The most defensible condition is finite all-layer robustness, rather than a product of raw weight norms. A sufficient, more diagnostic condition is bounded internal amplification throughout a perturbation tube. For smooth activations define

$$G_r(w) = \sup_{x, \|\eta\|_* \leq r} \left(1 + N \sum_{s<L} \|\nabla_{\eta^s} z_w(x;\eta)\|_2^2 \right)^{1/2} .$$

Integration along $t\eta$ gives output change at most $G_r(w)\|\eta\|_*$. Thus clean margin $> 1/2$ and $rG_r(w) \leq 1/3$ imply all-layer sign margin r. This uses actual downstream Jacobians, including cancellation and saturation, and must hold in a finite neighborhood, not only at the clean point.

If transition Jacobian norms on that tube are bounded by q_s, then

$$G_r(w)^2 \leq 1 + N\|v\|_2^2 \sum_{s=1}^{L-1} \prod_{j=s+1}^{L-1} q_j^2 .$$

Nonexpansive hidden transitions give $G_r \leq \sqrt{1 + (L-1)\kappa_N}$. More generally, $q_j \leq 1 + a_j$, $a_j \geq 0$, and $\sum_j a_j = O(\log L)$ give polynomial amplification. These are transparent sufficient restrictions, but raw spectral nonexpansiveness may exclude useful circuit implementations. Saturation can supply much better local certificates. Residual step-size bounds give related architectural examples, but residual networks are not silently substituted into the original tanh model.

Relevant primary literature includes:

- Wei–Ma's all-layer margin [3]: a closely related perturbation notion, generalization theory, and gradient-based margin optimization on practical image models.
- Wei–Ma's Lipschitz augmentation [4]: polynomial explicit depth dependence in measured hidden-state and interlayer-Jacobian quantities, including smooth activations. The remaining matrix norms need not be width-independent under our κ_N.
- Arora–Ge–Neyshabur–Zhang [5]: noise stability and compression in trained networks. Their random/compression-specific and example-local properties are weaker than our all-input adversarial condition.
- Wei–Chen–Ma [6]: robust circuit simulations already play a role in polynomial circuit-learning bounds for a different learner and parameter-margin objective. Our Gaussian-prior and width normalization need their own arguments.

These sources support naturalness and trainability. They do not prove that ordinary gradient descent yields an inverse-polynomial worst-input, simultaneous RMS robustness radius. The

distinction is material: IID noise and worst-case noise differ by a factor $\sqrt{N}$ in linearized response, and an all-input guarantee is much stronger than a typical-data guarantee. The attached note's referenced numerical scripts were not supplied; their reported experiments are not treated as independently reproduced here.

5.1 A small reproducible training probe

We trained 24 predetermined width-32 tanh networks on all 64 inputs of the six-bit cube: majority and parity on the first three bits, hidden depths $2, 4, 8, 16$, and three seeds. The optimizer was full-batch Adam, for 6000 steps; there were no later-layer biases. All fits succeeded, with minimum clean margins between 0.99607 and 0.99994. No failed fit or unusual seed was removed. This is a toy test of actual gradient-trained networks, not evidence about large practical models.

For the total noise norm (1), the following are medians over seeds of the worst radius across the entire cube. A sufficient radius is obtained from a nonlinear finite-neighborhood derivative bound; an attack radius is one at which an analytic perturbation or PGD gives a sign failure.

Task	Hidden depth	Sufficient radius	Found attack radius
Majority_3	2	.0751	.115
	4	.0361	.0790
	8	.0156	.0563
	16	.00644	.0566
Parity_3	2	.0576	.0938
	4	.0278	.0741
	8	.0119	.0597
	16	.00297	.0449

The analytic sufficient bound is valid, but numerical evaluation uses float64 SVD rather than interval-certified arithmetic. An unsuccessful attack is never a robustness certificate. One retained depth-16 parity seed has spectral product 7.38×10^6, sufficient local radius 4.99×10^{-5}, and a found attack at .0933; this is a large gap between available bounds, not a proof of robustness at .0933. The tested depth range and gaps do not establish polynomial scaling. Scripts, all-seed tables, saved weights, and raw data accompany this report.

A separate, subsequently predeclared stress cohort used hidden depths $32, 64$ with the same optimizer settings and three seeds for both tasks. All six depth-32 fits succeeded; only three of six depth-64 fits did. The three retained failures made 23, 31, and 30 errors on the 64 training inputs, so their robustness radius relative to the target labels is zero; this does not assert that robustness of their own clean predictions is zero. Among the successful depth-64 fits, sufficient sign radii ranged from 1.10×10^{-5} to 4.45×10^{-4}, while known attack radii ranged from .01195 to .120. This cohort is reported separately, not pooled with the original grid. It reinforces the limitation: neither optimization success nor an inverse-polynomial robustness law follows from ordinary training alone.

We also tested the exact support condition used in Theorem 4.4, recomputing S_N from every saved model and evaluating the analytic bound at $r = .01/\sqrt{1 + S_N}$. Of the original 24 models, 23 pass this sufficient test and one is unresolved. In the 12-model stress cohort, five pass, three violate the clean absolute margin requirement, and four are unresolved by that test. A stronger finite-radius certificate, derived in Appendix G, retains exact clean Jacobian products and bounds nonlinear Taylor remainders. Two independent implementations pass all five previously unresolved models, including the three successful depth-64 fits. Thus all 33 successful fits pass a sufficient test for the exact proposed support, and the three failed fits violate its clean absolute margin, subject

throughout to the floating-point qualification. The earlier and improved audits are both retained. These tests preserve each network's own clean labels, as required by the prior; a training error alone is not used as a robustness counterexample. For these 33 fits, the readout-reference radii imply the conservative activation-rounding guarantee above with 11–15 fractional bits, subject to the same numerical qualification.

6 What the necessity question still requires

The existing norm construction represents every truth table on q bits with

$$\kappa_N \le A\left[dq^2\log^2(2q) + (L-2)2^{2q/(L-2)}\right], \qquad N \ge 2^{q+1}.$$

At $q = d$, $L = d+2$, this is polynomial norm but 2^{2^d} functions. Appendix B strengthens it to global polynomial input Lipschitz constants. This invalidates a polynomial joint norm-capacity bound.

It does not establish the desired Gaussian posterior lower bound. Most truth tables have large target complexity. A large low-norm class does not show that the relevant competitors have lower rate than a particular simple target, nor that their integrated Gaussian mass is large enough after data. In particular, the common cost of transmitting a signal through many layers may cancel in target-relative rate comparisons.

There is also a positive statement for the *original* Gaussian prior. Appendix D proves that its globally minimum-energy Boolean functions are exactly $\{\pm 1, \pm x_1, \ldots, \pm x_d\}$, at every depth and width. The strong-prior function prior converges to the uniform distribution on these $2(d+1)$ functions. Consequently each of these targets has sample bound $O((\log d + \log(1/\delta))/\varepsilon)$ independent of depth and width; for uniform inputs, $O(\log(d/\delta))$ samples identify the target exactly in the limit. The sufficiently-small-variance threshold can depend on depth and width. This does not settle general circuit targets, but rules out using the constant target as a strong-prior Gaussian counterexample merely by invoking the huge low-norm class. The common minimum rate is also quantitatively understood: writing $D_0 = \max\{d, 4\}$, it lies between $L(d/4)^{1/L}$ and $(L+12)(D_0/4)^{1/L}$. Thus it is $L + O_d(1)$ at large depth, even though the corresponding sample bound is independent of depth.

Appendix E further separates energy, robustness, and statistical difficulty. Energy $L+B$ bounds the total squared error of best rank-one approximations to the internal matrices by $B - \log(d/4)$. Yet a constant-function network can have energy at most $K_*(4, L) + 76$ and be erased by hidden RMS noise as small as $L^{-1}e^{-\sqrt{L}/4}$. Thus bounded excess energy does not imply polynomial robustness of every cheap realization. Conversely, appending layers efficiently shows that every fixed representable function has infimum energy $L + O(1)$ as depth grows, with the offset depending on the function, input dimension, and width. Exponential growth of minimum target energy cannot itself explain a depth-dependent statistical lower bound.

The distinction can be made exact. A *different* prior assigning equal mass to all truth tables in the construction needs $\Omega(2^d)$ examples even for the constant target: every unseen input has posterior probability $1/2$, so its $> 2/3$ confidence-failure fraction is at least $1 - n/2^d$. Small parameter neighborhoods can replace the atoms while preserving their cube labels and polynomial energy bounds, without changing this statistical conclusion. This does not assert stability of the global input-Lipschitz bound under arbitrary parameter perturbations. It is a norm-supported-prior counterexample, not a counterexample to the centered independent Gaussian prior.

Conversely, a circuit-description prior mixed with a positive full-support Gaussian component can retain fragile networks in support and still give polynomial-depth sample bounds through target

prior mass. Thus hard robustness support is not logically necessary for Bayesian learning in general. The precise unresolved question concerns the particular Gaussian learner.

The empirical next step is to measure all-layer margins or sound finite-tube certificates across depth and width, keeping random-noise tests, attacks, and certificates separate. The theoretical next step is to settle the unrestricted Gaussian case, by a general polynomial-depth bound or a genuine simple-target posterior lower bound. Replacing all-input robustness by typical-input robustness would be valuable, but needs a new statistical argument; arbitrary exceptional sets can otherwise memorize the training data.

A Robust Gaussian target mass

This appendix supplies the denominator used in Theorems 4.2 and 4.4, for every $N \geq C$ and every $0 < t \leq 1$. Take balanced clone groups of sizes k_a, so $N/k_a \leq 2C$, and let P project onto group-constant vectors. Every unused circuit slot is filled by a Boolean constant, so the ideal wire vector $b^s(x)$ lies in the range of P and has norm $\sqrt{N}$. Put

$$e_* = (16\sqrt{2C})^{-1}, \quad \tau_0 = e_*/4, \quad b_L = \sqrt{16\log 9 + 8\log(4L)},$$

$$A = \max\{32b_L, 2\log(8/e_*)\}, \qquad \rho = \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}.$$

Use tanh gate gain A and readout coefficient 2. The gain need not be bounded by logical circuit width; it can be accommodated by polynomial padding if invoking the "weights bounded by width" complexity convention.

Allow any $0 < t \leq 1$. The standardized coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv have common standard deviation t; the exponent-γ case is $t = N^{(1-\gamma)/2}$. Restrict every first-layer coordinate, every internal row's projection onto each normalized clone-group direction, and every readout coordinate to an interval of radius ρt around its clone value. Then

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t\sqrt{C}, \quad E^s P = 0.$$

The free Gaussian bulk E^s is independent of the cylinder. A 1/4-net Gaussian operator bound makes $\|E^s\|_{\mathrm{op}} \leq b_L t$ simultaneously with conditional probability at least 1/2.

We check robustness to a separate per-layer RMS budget τ_0, which is stronger than the total-budget requirement for $r \leq \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \leq e_*$, each clone-group mean has error at most 1/16, so the teacher part of every intended signed preactivation is at least $A/2$. For every real u and $\epsilon \in \{-1, 1\}$,

$$\epsilon a \geq A/2 \quad \Longrightarrow \quad |\tanh(a+u) - \epsilon| \leq 2e^{-A/2} + 8|u|/A.$$

This is a global secant bound: for $|u| < A/4$ use saturation, and otherwise use the trivial bound 2. It handles concentrated adversarial noises. Since $E^s b^{s-1} = 0$ and noisy states have norm at most $(1+\tau_0)\sqrt{N}$,

$$e_s \leq 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \leq e_*.$$

The first layer starts the invariant because its coordinatewise preactivation error is at most $\rho(d+1) \leq A/4$. The readout has noisy signed value at least $2 - 1/8 - \rho(1+\tau_0) - \tau_0 > 1$.

There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. Their squared centers sum to at most NB'_0, with

$$B'_0 = 9dA^2 + 18(L-2)CA^2 + 8C.$$

The elementary Gaussian interval bound and the independent operator event give

$$-\log \mu_{N,t}(\mathcal{R}_{r,N,f}) \le \frac{N}{t^2}\left[B_0' + q\log\frac{e\sqrt{2\pi}}{2\rho} + \log 2\right] = DN/t^2.$$

Here D is jointly polynomially bounded in d, C, L. Exponentially small coordinate precision would in any case cost only its logarithm in this negative log-mass estimate; it should not be replaced unnecessarily by its reciprocal.

B Input robustness does not control the depth-dependent norm capacity

The following statement concerns norm-class capacity, not the posterior of the independent Gaussian weight prior.

Proposition B.1. *For every integer $d \ge 2$, put $N = 2^{d+1}$ and $L = d+3$. Every $f : \{-1,1\}^d \to \{-1,1\}$ has a width-N, depth-L tanh network, with a linear readout and no biases except the constant input coordinate, whose real-valued output $z : \mathbb{R}^d \to \mathbb{R}$ satisfies*

$$z(x) = f(x) \quad (x \in \{-1,1\}^d), \qquad \kappa_N(z) \le 100d^3\log^2(16d), \qquad \mathrm{Lip}(z) \le 300d^{3/2}\log^2(16d).$$

Here κ_N is the weighted squared parameter norm from the main paper. In particular the correct signed output remains at least $3/4$ under every Euclidean input perturbation of radius $[1200d^{3/2}\log^2(16d)]^{-1}$ around a Boolean vertex. Nevertheless this witness fails the hidden-activation robustness condition at RMS radius

$$\tau_d = \frac{3\sqrt{2}}{4}\,2^{-d}.$$

Thus polynomial bounds on weighted norm and even global input Lipschitz constant do not give a norm-class counting bound polynomial jointly in depth and the displayed parameters.

Proof. Put $M = 2^d$, $m = d-1$, $s = 3/4$, $a = \frac{1}{2}\log(16d)$, $B = \log(4d)$, $c = \tanh B$, and $b = \tanh 1$. For $p \in \{-1,1\}^d$ define

$$T_p(u) = \frac{1 + \tanh(a(p\cdot u - (d-1)))}{2}.$$

The $M \times M$ matrix $T = (T_p(x))_{x,p}$, evaluated on the Boolean cube, satisfies

$$\|T - I\|_\infty \le e^{-2a} + e^{2a}\left[(1+e^{-4a})^d - 1\right] < \frac{1}{4}.$$

Indeed the first term bounds its diagonal defect and the second sums the positive off-diagonal terms by Hamming distance. Consequently $v = T^{-1}f$ satisfies $\|v\|_\infty \le 4/3$. Write $F(u) = \sum_p v_p T_p(u)$.

The first hidden layer contains $r = \lfloor N/(d+1)\rfloor$ copies of $\tanh(Bx_i)$ for each $i = 1,\dots,d$, and r copies of the constant c; any unused neurons are zero. The next layer has M detector neurons. For detector p, give each copy of coordinate i weight $ap_i/(rc)$ and each constant copy weight $-a(d-1)/(rc)$. This computes $\tanh(a(p\cdot u(x) - (d-1)))$, where

$$u_i(x) = \frac{\tanh(Bx_i)}{c}.$$

Its other M neurons compute the constant b, using weight $1/(rc)$ from each of the r constant predecessors.

Set

$$t_0 = s2^{-(d-2)/2}, \qquad R = 2^{(d-2)/(2(d-1))}, \qquad t_j = t_0 R^j \quad (0 \le j \le m).$$

The third hidden layer has one active neuron with weights $a_p = \frac{1}{2}\operatorname{atanh}(t_0)v_p$ from the detectors and common weight $(\sum_p a_p)/(Mb)$ from the constant neurons. It computes $\tanh(\operatorname{atanh}(t_0)F(u(x)))$. For each of the next m hidden layers put $n_j = \lceil N^{j/m} \rceil$ active neurons, where $n_0 = 1$. Each has incoming weight $\operatorname{atanh}(t_j)/(n_{j-1}t_{j-1})$ from every active predecessor. Finally use readout weight $1/(Ns)$ on each last-layer neuron. Since u fixes the Boolean vertices and $F(x) = f(x)$ there, induction shows that the active neurons of stage j equal $t_j f(x)$ and the output is $f(x)$. There are $3 + m = d + 2$ hidden layers, hence depth $L = d + 3$.

For completeness, the norm accounting has no hidden exponential factor. The first layer costs at most dB^2. Since $r \ge N/[2(d+1)]$, the second layer costs at most

$$\frac{d+1}{c^2}\big[a^2(d + (d-1)^2) + 1\big] \le 2(d+1)\big[a^2(d^2 - d + 1) + 1\big].$$

The third-layer cost is at most $(1 + b^{-2})(\operatorname{atanh}(s)/s)^2 < 5$, using $Mt_0^2 = 9/4$. Each amplification layer costs at most

$$2\Big(\frac{\operatorname{atanh}(s)}{s}\Big)^2 N^{1/m} R^2 = 2\Big(\frac{\operatorname{atanh}(s)}{s}\Big)^2 2^{2+1/(d-1)} < 27,$$

and the readout costs $s^{-2} = 16/9$. These bounds imply the stated $100d^3 \log^2(16d)$ bound with ample slack.

We next bound input sensitivity on all of $\mathbb{R}^d$. One has $c^{-1} \le 1 + 1/(7d^2)$ and $\operatorname{Lip}(u) \le B/c < 2B$. For $|u_i| \le c^{-1}$,

$$\begin{aligned}\sum_p e^{2a(p\cdot u - (d-1))} &\le 16d \exp\Big(\frac{1}{256d} + 2ad(c^{-1} - 1)\Big) \\ &\le 16d \exp\Big(\frac{1}{256d} + \frac{\log(16d)}{7d}\Big) < 21d.\end{aligned}$$

Using $\operatorname{sech}^2(t) \le 4e^{2t}$ and $\|v\|_\infty \le 4/3$ gives $\|\nabla F(u)\|_2 \le 56ad^{3/2}$ throughout that box.

The remainder of the network is a scalar map H, so $z = H \circ F \circ u$. By $|\tanh'| \le 1$ and telescoping the scale factors,

$$\operatorname{Lip}(H) \le \prod_{j=0}^{m} \frac{\operatorname{atanh}(t_j)}{t_j}.$$

For $0 < t \le s$, the power series gives $\log(\operatorname{atanh}(t)/t) \le t^2/[3(1 - s^2)] = 16t^2/21$. If $d \ge 3$, then $R^2 \ge \sqrt{2}$ and $\sum_j t_j^2 \le s^2/(1 - 1/\sqrt{2})$, hence

$$\operatorname{Lip}(H) \le \exp\left(\frac{3}{7(1 - 1/\sqrt{2})}\right) < 5.$$

The case $d = 2$ has two values $t_j = s$ and satisfies the same bound. Multiplying the three Lipschitz bounds yields

$$\operatorname{Lip}(z) \le 560aBd^{3/2} \le 280d^{3/2}\log^2(16d) < 300d^{3/2}\log^2(16d).$$

The claimed input-perturbation guarantee follows from $z(x) = f(x)$.

Finally, at any Boolean input the third hidden layer has just one nonzero coordinate, $t_0 f(x)$. An additive post-activation perturbation $-t_0 f(x)$ in that coordinate sets the layer to zero. All subsequent hidden activations and the output are then zero. Its Euclidean norm is t_0, and

$$\frac{t_0}{\sqrt{N}} = \frac{3\sqrt{2}}{4} 2^{-d}.$$

Thus the internal robustness condition fails at that RMS radius. The 2^{2^d} truth tables give the counting conclusion. □

Scope. This rules out polynomial-depth *norm-class counting* under polynomial input-Lipschitz bounds, and gives an explicit internal-fragility mechanism. It does not prove a sample lower bound for a simple target under the original independent Gaussian parameter prior: that would require a posterior mass or minimum-energy comparison specific to that prior.

C Polynomial-overhead robustification

Proposition C.1. *Suppose a width-C, depth-L tanh network, with every weight bounded in absolute value by C, implements $f : \{-1,1\}^d \to \{-1,1\}$ with strict output margin $1/2$. Then f has a tanh implementation of width and depth polynomial in (d, C, L) that tolerates simultaneous adversarial hidden-layer activation perturbations of inverse-polynomial RMS size. More precisely, there is a fan-in-two Boolean circuit for f of size $S = \mathrm{poly}(d, C, L)$. Put*

$$W = \max\{8, d + S + 1\}, \qquad D = S + 2.$$

There is a width-W, depth-D tanh network, all of whose weights have absolute value at most 8, such that every sequence of post-activation injections with

$$\|\zeta^s\|_2 \le \frac{1}{8} \quad (1 \le s < D), \qquad |\zeta^{\mathrm{out}}| \le \frac{1}{8\sqrt{W}}$$

preserves signed output margin strictly greater than $5/8$, on every Boolean input. Its weighted norm is at most

$$\frac{64d(d+1)}{W} + 192SW + W.$$

At every width $N \ge W$, cloning preserves this guarantee at RMS radius $1/(8\sqrt{2W})$ and increases weighted norm by at most a factor of two. The construction allows polynomial depth and width overhead; it makes no claim at the original depth L or every original width $N \ge C$.

Proof. We first replace the analog computation by a finite Boolean computation. Round every parameter to absolute precision $\eta = 2^{-p}$. Evaluate each hidden tanh to precision η and clip its approximation into $[-1, 1]$; evaluate the readout to the same precision. If E_s is the maximum coordinate error after layer s, the first hidden layer satisfies $E_1 \le (d+2)\eta$, and the subsequent layers and readout satisfy

$$E_s \le C^2 E_{s-1} + (C+1)\eta.$$

This follows by writing each affine error as the original weights times activation errors, plus weight errors times bounded approximate activations. The clipping operation does not increase the approximation error. It suffices to choose

$$2^{-p}\,[d + 2 + L(C+1)](C^2+1)^{L-1} \le \frac{1}{8}.$$

Thus $p = O(L\log(C+1) + \log(d+L+1))$ bits suffice, and the final sign is unchanged. The rounded parameters are hardwired constants; no efficient description of the original real numbers is required.

Here is an elementary polynomial-cost tanh evaluation algorithm. For a dyadic argument t with $|t| \geq p+2$, return its sign; the error is less than 2^{-p}. Otherwise set $x = 2|t| \leq 2(p+2)$ and approximate e^{-x} by its degree $m = 16(p+2)$ Taylor polynomial. Its error is at most

$$\frac{x^{m+1}}{(m+1)!} \leq \left(\frac{ex}{m+1}\right)^{m+1} \leq (e/8)^{m+1} < 2^{-p-4}.$$

The Taylor sum can be evaluated exactly using the common denominator $2^{bm}m!$, where b is the number of fractional bits of x. All integers involved have polynomial bit length, even though some Taylor terms are large and cancel. Round and clip the answer into $[0,1]$, then use $(1-u)/(1+u)$ and oddness to obtain $\tanh t$. This rational map has derivative of magnitude at most 2 on $[0,1]$, so a constant number of guard bits gives the required accuracy. Integer addition, multiplication, comparison, and division have polynomial-size Boolean circuits. Applying this algorithm throughout the rounded network gives a Boolean circuit of size $S = \mathrm{poly}(d, C, L)$ for f.

We now simulate that circuit by noisy tanh gates. Express bits as ± 1, and carry a constant $+1$ channel. Use

$$\begin{aligned} &\mathrm{COPY}(a) = \tanh(8a), && \mathrm{NOT}(a) = \tanh(-8a), \\ &\mathrm{AND}(a,b) = \tanh(8(a+b-c)), && \mathrm{OR}(a,b) = \tanh(8(a+b+c)), \end{aligned}$$

where c is the constant channel, itself updated by $\tanh(8c)$. Simplify gates with repeated or constant inputs beforehand, so each displayed nonzero edge has magnitude 8. If all incoming signals are within $1/4$ of their ideal values, the signed affine expression for AND and OR is at least $1 - 3/4 = 1/4$; COPY and NOT have larger margin. Consequently each gate output is within $1 - \tanh 2 < 1/8$ of its ideal value. Adding any coordinate perturbation of magnitude at most $1/8$ leaves the error strictly below $1/4$. This invariant holds at arbitrarily many layers, with no accumulated error.

Use one Boolean gate per hidden layer and copy all previously computed signals forward. The first hidden layer initializes the input signals and constant from $\tanh(8x_i)$, including $x_0 = 1$. The W channels accommodate all inputs, gate outputs, and the constant. Unused channels have zero outgoing weights until used. Each hidden-layer Euclidean noise budget $1/8$ bounds every coordinate injection by $1/8$, so the invariant applies. Read out the desired final signal with coefficient one. It has signed value greater than $3/4$; the output injection is at most $1/8$, leaving signed margin greater than $5/8$. The first layer has $d+1$ nonzero edges, each of the S gate layers has at most $3W$ nonzero edges, and the readout has one edge. This gives the stated weighted norm.

Finally, at width $N \geq W$ use $k = \lfloor N/W \rfloor$ copies of each hidden neuron, divide outgoing weights by k, and leave the remaining channels dead. Clone-group means obey exactly the original computation with averaged post-activation injections. Their Euclidean noise norm is at most

$$\frac{1}{8\sqrt{2W}}\sqrt{\frac{N}{k}} \leq \frac{1}{8},$$

because $N/k \leq 2W$. Thus the same invariant and margin hold. The first-layer norm does not increase, internal-layer norm is unchanged, and the weighted readout norm increases by at most two. □

Interpretation. Inverse-polynomial internal robustness does not prevent representation of any Boolean function already having a bounded-weight tanh implementation, provided polynomial

increases in depth and width are allowed. This is an expressivity statement, not a claim that gradient descent produces the constructed digital simulation. The point is that exponentially small analog precision costs only polynomially many digital bits.

D A depth-independent Gaussian result for the simplest targets

The norm-capacity obstruction does not automatically make even the simplest targets hard for the specified Gaussian prior. In fact constants and dictators have a depth-independent sample bound in the strong-prior limit.

Theorem D.1. *Fix $d \geq 1$, $N \geq 1$, and $L \geq 2$, with tanh hidden activations, the constant input coordinate, and a linear readout as in the main paper. Let*

$$\mathcal{M}_d = \{+1, -1, +x_1, -x_1, \ldots, +x_d, -x_d\}.$$

Among all Boolean functions implemented with the all-input strict margin $1/2$, the functions of smallest weighted norm are exactly $\mathcal{M}_d$. Their common infimum norm value depends on d, L but not on N.

Consequently, for every $f \in \mathcal{M}_d$ and every $\varepsilon, \delta \in (0,1)$, there is $\sigma_0 > 0$ such that the original globally margin-conditioned Gaussian posterior has posterior-draw risk at most ε, with probability at least $1 - \delta$, whenever

$$0 < \sigma < \sigma_0, \qquad n \geq \frac{2}{\varepsilon}\left(\log(2(d+1)) + \log\frac{1}{\delta}\right).$$

The guarantee holds for arbitrary input distributions on the Boolean cube. The sample bound is independent of depth and learner width. The variance threshold may depend on d, L, N, ε, f; no uniform threshold over depths or widths is asserted.

Proof. First, the function $Q(u) = \tanh^2\sqrt{u}$ is strictly increasing and strictly concave for $u \geq 0$. To see concavity, put $t = \sqrt{u} > 0$ and $y = \tanh t$. A direct derivative computation gives

$$Q''(u) = \frac{1 - y^2}{2t^3}\left[t(1 - 3y^2) - y\right] < 0.$$

Indeed $y - t(1 - y^2) > 0$ for $t > 0$, since it vanishes at zero and has derivative $2ty(1 - y^2) > 0$; hence the bracket is less than $-2ty^2$. The continuous extension at zero preserves concavity. Jensen's inequality therefore implies, for every $v \in \mathbb{R}^N$,

$$\frac{\|\tanh v\|_2}{\sqrt{N}} \leq \tanh\left(\frac{\|v\|_2}{\sqrt{N}}\right). \tag{4}$$

For any network weights define

$$a = \frac{\|W^1\|_F}{\sqrt{N}}, \qquad b_s = \|W^s\|_F \ (2 \leq s < L), \qquad c = \sqrt{N}\|v\|_2.$$

Their weighted squared norm is

$$\kappa_N = da^2 + \sum_{s=2}^{L-1} b_s^2 + c^2.$$

Define the scalar recursion $r_1 = \tanh a$, $r_s = \tanh(b_s r_{s-1})$ for $2 \leq s < L$, and let $F(a, b, c) = c r_{L-1}$. Let $K_*(d, L)$ be the minimum of the displayed cost over nonnegative $(a, b_2, \ldots, b_{L-1}, c)$ with $F \geq 1/2$.

The minimum exists: the constraint is closed and nonempty and the cost is coercive. Every feasible tuple has all entries positive. At a minimizer, $F = 1/2$, since otherwise decreasing c lowers the cost.

Every tuple is realized by a constant-channel network: all first-layer rows have weight a on the constant input and zero elsewhere; every entry of W^s is b_s/N; and every readout entry is c/N. Its output is the constant F and its norm is the stated cost, at every width. Moving the first-layer weight from the constant input to input x_i gives output Fx_i by oddness of tanh; negating the readout gives the negative functions. Thus all members of $\mathcal{M}_d$ have infimum norm at most K_*, with strict margins obtained by an arbitrarily small increase of the readout.

Conversely, consider a network with $|z_w(x)| \geq 1/2$ at every Boolean input. For uniform x on the Boolean cube and $\bar{x} = (1, x)$,

$$\mathbb{E}_x \frac{\|W^1 \bar{x}\|_2^2}{N} = \frac{\|W^1\|_F^2}{N} = a^2.$$

There is therefore an x_* with $\|W^1 \bar{x}_*\|_2/\sqrt{N} \leq a$. Apply (4) at each hidden layer, the inequality $\|Wh\|_2 \leq \|W\|_F \|h\|_2$ at each internal affine map, and Cauchy–Schwarz at the readout. These give

$$\frac{1}{2} \leq |z_w(x_*)| \leq F(a, b, c).$$

Thus its norm is at least K_*. In particular, the members of $\mathcal{M}_d$ have exactly the global infimum norm K_*.

We check the equality case to exclude all other Boolean functions. Suppose a globally margin-$1/2$ network has norm exactly K_*. Its associated scalar tuple minimizes the scalar problem, so $F = 1/2$. All parameters in that tuple are positive, so the scalar recursion is strictly increasing in the first preactivation RMS. Choose x_* minimizing that RMS. If its value were strictly less than a, then the preceding chain would yield $|z_w(x_*)| < 1/2$. Hence the minimum RMS equals its quadratic mean a, so

$$\frac{\|W^1 \bar{x}\|_2}{\sqrt{N}} = a \quad \text{for every Boolean } x.$$

For every input, the entire output bound is consequently an equality: $|z_w(x)| = F = 1/2$. In particular the first hidden vector has the fixed nonzero norm $\sqrt{N} \tanh a$. If $L = 2$, equality in readout Cauchy–Schwarz says that all first hidden vectors are collinear with the same nonzero readout vector. If $L \geq 3$, equality in $\|W^2 h^1(x)\|_2 \leq \|W^2\|_F \|h^1(x)\|_2$ says the same: every nonzero row of W^2 is collinear with $h^1(x)$, and $W^2 \neq 0$. Thus, fixing any reference input x_0, there is $\epsilon(x) \in \{-1, 1\}$ with

$$h^1(x) = \epsilon(x) h^1(x_0).$$

Oddness and injectivity of tanh imply $W^1 \bar{x} = \epsilon(x) W^1 \bar{x}_0$. Choose a nonzero coordinate of $W^1 \bar{x}_0$; the corresponding first-layer affine form therefore has constant nonzero absolute value on the cube. An affine form $\alpha_0 + \sum_i \alpha_i x_i$ with constant squared value can have only one nonzero coefficient: expanding its square forces $\alpha_0 \alpha_i = 0$ and $\alpha_i \alpha_j = 0$ for $i \neq j$. Therefore ϵ is a signed constant or dictator. Every remaining layer is odd, so $z_w(x) = \epsilon(x) z_w(x_0)$, proving that its Boolean function belongs to $\mathcal{M}_d$.

The strict-margin infimum causes no missing equality case. For each representable Boolean function g, replace strict constraints by $g(x) z_w(x) \geq 1/2$. The relaxed feasible set is closed and its minimum norm is attained by coercivity in the finite-dimensional parameter space. Its minimum equals the strict infimum: scaling the readout by $1 + \eta$ gives strict feasibility with norm converging to the relaxed value as $\eta \downarrow 0$. Consequently every representable $g \notin \mathcal{M}_d$ has $\kappa_N(g) > K_*$.

Finally, the small-variance Gaussian calculation in Section 4.2, applied to the open strict-margin implementation sets $U_{N,g}$, gives

$$\frac{\mu_{N,\sigma}(U_{N,g})}{\mu_{N,\sigma}(U_{N,f})} \longrightarrow 0 \quad \text{when } \kappa_N(g) > \kappa_N(f).$$

There are finitely many Boolean functions. Therefore, for $f \in \mathcal{M}_d$,

$$\frac{\pi_{N,\sigma}(\mathcal{M}_d^c)}{\pi_{N,\sigma}(f)} \longrightarrow 0.$$

Choose σ_0 so that this ratio is at most $\varepsilon/2$. At the displayed sample size, a union bound shows that, with probability $1-\delta$, every consistent member of $\mathcal{M}_d$ has error at most $\varepsilon/2$. The posterior mass outside $\mathcal{M}_d$ is at most $\pi_{N,\sigma}(\mathcal{M}_d^c)/\pi_{N,\sigma}(f) \le \varepsilon/2$, because f is always consistent. Averaging errors proves the claimed posterior risk guarantee. □

Scope. This is a result for the actual independent Gaussian prior in the strong-prior limit, with the all-input margin conditioning retained. It does not settle arbitrary circuit-simple targets, the specified nonzero variances in the $\gamma \ge 1$ papers, or a width/depth-uniform variance threshold. It shows specifically why a uniform-truth-table low-norm prior counterexample with a constant target cannot simply be transferred to this Gaussian prior.

Corollary D.2. *For every fixed d, N, L, the globally margin-conditioned Gaussian function prior converges, as $\sigma \downarrow 0$, to the uniform distribution on $\mathcal{M}_d = \{\pm 1, \pm x_1, \ldots, \pm x_d\}$. For data labeled by any $f \in \mathcal{M}_d$, the posterior converges to the uniform distribution on the members of $\mathcal{M}_d$ consistent with the data. In fact this posterior convergence is uniform over all such labeled samples, including arbitrary sample sizes. For uniform IID inputs, if*

$$n \ge \log_2 \frac{2d+1}{\delta},$$

then with probability at least $1-\delta$ the limiting posterior is the point mass at f.

Proof. The norm theorem shows that the prior mass outside $\mathcal{M}_d$ tends to zero relative to the mass of any one member. Their masses are exactly equal at every positive variance. Indeed, swapping columns 0 and i in the first weight matrix preserves its independent Gaussian law. Let T_i be the involution of the Boolean cube defined by

$$(T_i x)_i = x_i, \qquad (T_i x)_j = x_i x_j \quad (j \ne i).$$

Since the entire network, as a function of its augmented input, is odd, this column swap transforms its output into

$$z_{w'}(x) = x_i z_w(T_i x).$$

It therefore bijects the strict-margin implementation sets for $+1$ and $+x_i$. Negating the readout preserves its Gaussian law and supplies the negative functions. Thus all $2(d+1)$ implementation sets have equal mass.

Conditioning on any sample labeled by $f \in \mathcal{M}_d$ retains at least f. If π is the function prior, the posterior mass outside $\mathcal{M}_d$ is bounded by $\pi(\mathcal{M}_d^c)/\pi(f)$, independently of the sample and its size. Within $\mathcal{M}_d$, consistency conditioning is exactly uniform. This proves the posterior convergence claim. Under uniform inputs, each distinct $g \in \mathcal{M}_d$ disagrees with f on at least half the cube. Hence its probability of surviving n labels is at most 2^{-n}. A union bound over the at most $2d+1$ competitors proves the last assertion. □

Proposition D.3 (The common depth cost)**.** *Let $K_*(d, L)$ be the scalar minimum from the preceding theorem and put $D = \max\{d, 4\}$. Then*

$$L(d/4)^{1/L} \;\le\; K_*(d, L) \;\le\; (L + 12)(D/4)^{1/L}.$$

In particular, for every fixed input dimension, $K_*(d, L) = L + O_d(1)$ as $L \to \infty$. More precisely,

$$\log(d/4) \le \liminf_{L\to\infty}(K_*(d, L) - L) \le \limsup_{L\to\infty}(K_*(d, L) - L) \le 12 + \log(D/4).$$

Proof. Since $\tanh t \le t$ for $t \ge 0$, scalar feasibility implies $ca\prod_{s=2}^{L-1} b_s \ge 1/2$. The product of the L weighted cost terms $da^2, b_2^2, \dots, b_{L-1}^2, c^2$ is at least $d/4$. The arithmetic–geometric mean inequality proves the lower bound.

For the upper bound set

$$q = (D/4)^{1/(2L)} \ge 1, \qquad k_s = \min\{s, L - s\}, \qquad r_s = \frac{q^s}{\sqrt{D}\, k_s} \quad (1 \le s < L).$$

Then $0 < r_s \le 1/(2qk_s) \le 1/2$ and $r_{L-1} = 1/(2q)$. Take

$$a = \operatorname{atanh}(r_1), \qquad b_s = \frac{\operatorname{atanh}(r_s)}{r_{s-1}}, \qquad c = q.$$

This realizes the hidden amplitudes r_s exactly and has output $1/2$. For $0 < t \le 1/2$, the power series gives

$$\frac{\operatorname{atanh} t}{t} \le 1 + \frac{t^2}{3(1 - t^2)} \le 1 + \frac{4}{9}t^2, \qquad \left(\frac{\operatorname{atanh} t}{t}\right)^2 \le 1 + t^2.$$

(The last step uses $t^2 \le 1/4$.) Hence

$$da^2 \le dr_1^2 + dr_1^4 \le q^2 + \frac{q^4}{D} \le q^2 + \frac{1}{4}, \qquad c^2 = q^2.$$

Put $R_s = k_{s-1}/k_s$ for $2 \le s < L$. The ratios occur in reciprocal pairs $k/(k+1), (k+1)/k$, with at most one additional ratio equal to one. Each pair contributes at most $4/k^2$ above two, since

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 = \left(\frac{k+1}{k} - \frac{k}{k+1}\right)^2 \le \frac{4}{k^2}.$$

Therefore $\sum_{s=2}^{L-1} R_s^2 \le L - 2 + 4\sum_{k\ge1} k^{-2} < L - 2 + 7$. Moreover $R_s \le 2$ and $q^{2s+2} \le q^{2L} = D/4$, so

$$\begin{aligned}\sum_{s=2}^{L-1} b_s^2 &\le q^2 \sum_{s=2}^{L-1} R_s^2 + \sum_{s=2}^{L-1} q^2 R_s^2 r_s^2 \\ &\le q^2(L - 2 + 7) + \sum_{s=2}^{L-1} \frac{1}{k_s^2} < q^2(L - 2 + 7) + 4.\end{aligned}$$

Adding the endpoint costs gives $K_*(d, L) \le q^2(L+7) + 17/4 \le q^2(L+12)$. The argument also applies at $L = 2$, when the internal sums are empty. Finally, expand $Le^{\log(d/4)/L}$ and $(L + 12)e^{\log(D/4)/L}$ for fixed d to obtain the asymptotic statements. □

E What excess Gaussian energy does and does not control

Write a width-N, depth-L tanh network as $h^1 = \tanh(W^1\bar{x})$, $h^s = \tanh(W^s h^{s-1})$ for $2 \le s < L$, and $z = v^T h^{L-1}$, where $\bar{x} = (1, x)$ and $x \in \{-1, 1\}^d$. Its weighted Gaussian energy is

$$\kappa_N = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

The following facts concern this actual energy. They do not establish a sample-complexity lower bound for the unrestricted Gaussian posterior.

Proposition E.1 (Excess energy controls total deviation from rank one)**.** *Suppose $|z(x)| \ge 1/2$ on the whole Boolean cube. Let R_s be a best rank-one Frobenius approximation to W^s, and let $\rho_s = \|W^s\|_F^2/\|W^s\|_{\mathrm{op}}^2$ be its stable rank. Then*

$$\sum_{s=2}^{L-1} \|W^s - R_s\|_F^2 \le \kappa_N - L(d/4)^{1/L}, \tag{5}$$

$$\sum_{s=2}^{L-1} \log \rho_s \le L \log(\kappa_N/L) - \log(d/4). \tag{6}$$

In particular, if $\kappa_N \le L + B$, both right-hand sides are at most $B - \log(d/4)$.

Proof. Set $a = \|W^1\|_F/\sqrt{N}$, $c = \sqrt{N}\|v\|_2$, and $p_s = \|W^s\|_{\mathrm{op}}$. Cube averaging supplies an input with $\|W^1\bar{x}\|_2/\sqrt{N} \le a$. Tanh contraction gives $1/2 \le ac\prod_s p_s$. Thus the product of the L numbers $da^2, p_2^2, \ldots, p_{L-1}^2, c^2$ is at least $d/4$, and their sum is at least $L(d/4)^{1/L}$. Subtract this sum from κ_N and use $\|W^s - R_s\|_F^2 = \|W^s\|_F^2 - p_s^2$ to obtain (5). Alternatively, the product of the Frobenius energy terms is at least $(d/4)\prod_s \rho_s$ and at most $(\kappa_N/L)^L$, which proves the logarithmic bound. Finally use $e^u \ge 1 + u$ and $\log(1 + u) \le u$. □

Thus bounded excess permits only a bounded total squared residual after replacing internal matrices by rank-one matrices. This does not justify replacing those matrices in the computation: the suffix can amplify the replacement errors. The next example shows that even a constant excess allowance does not imply inverse-polynomial hidden-state robustness.

Proposition E.2 (Bounded excess permits a fragile scalar transmission)**.** *For $d = 4$, every even $L \ge 4$, and every $N \ge 1$, there is a network with $z(x) = 1$ for all inputs and*

$$\kappa_N \le L + 76 \le K_*(4, L) + 76,$$

whose output can be made zero by a perturbation at one hidden layer of normalized RMS at most $L^{-1}e^{-\sqrt{L}/4}$. Here $K_*(4, L)$ is the globally minimum margin-$1/2$ energy characterized in the main text.

Proof. Put $n = L/2$, $h = L^{-1/2}$, $k_s = \min(s, L - s)$, and

$$r_s = \frac{e^{-h(k_s-1)}}{2k_s}, \qquad 1 \le s < L.$$

Give each first-layer row the sole constant-input weight $\operatorname{atanh}(1/2)$. Let every entry of W^s be b_s/N, where $b_s = \operatorname{atanh}(r_s)/r_{s-1}$, and let $v_i = 2/N$. The hidden vector is exactly $r_s\mathbf{1}$, so the output is $2r_{L-1} = 1$ and the energy is

$$4\operatorname{atanh}(1/2)^2 + \sum_{s=2}^{L-1} b_s^2 + 4.$$

For completeness, the following loose constants are uniform in depth. The ratios r_s/r_{s-1} occur in reciprocal pairs t_k, t_k^{-1}, where $t_k = (1+1/k)e^h$ and $1 \le k < n$. Since $h \le 1/2$,

$$t_k - t_k^{-1} \le e^h(2/k + 2h), \qquad t_k^2 + t_k^{-2} - 2 \le 8e^{2h}(k^{-2} + h^2).$$

Using $e^{2h} < 3$, $\sum_{k\ge1} k^{-2} < 2$, and $(n-1)h^2 \le 1/2$ gives $\sum_s (r_s/r_{s-1})^2 < L - 2 + 60$. Also $\max_s (r_s/r_{s-1})^2 < 12$ and $\sum_s r_s^2 < 1$. For $0 < t \le 1/2$ the power series gives $(\operatorname{atanh}(t)/t)^2 \le 1 + t^2$. Consequently

$$\sum_s b_s^2 < L - 2 + 72, \qquad 4\operatorname{atanh}(1/2)^2 \le 5/4, \qquad \kappa_N < L + 76.$$

The baseline lower bound gives $K_*(4, L) \ge L$. Finally, adding $-r_n\mathbf{1}$ after the middle activation zeros that state and hence every later state. Its normalized RMS is

$$r_n = L^{-1}\exp\left(-\frac{L/2-1}{\sqrt{L}}\right) \le L^{-1}e^{-\sqrt{L}/4}.$$

□

This example represents a statistically easy target: the earlier theorem proves depth-independent strong-prior learning for constants. It therefore separates fragility of a cheap parameter realization from difficulty of learning its Boolean function. It does not show that the fragile realization is a minimum-energy representative.

Proposition E.3 (Additional depth has asymptotically unit energy cost)**.** *Suppose a width-N, depth-L_0 network realizes f with margin at least $1/2$. Write $c^2 = N\|v\|_2^2 > 0$ and $E_{\mathrm{pre}} = \kappa_N - c^2$. For every integer $t \ge 1$, there is a width-N, depth-$(L_0 + t)$ realization of the same function, with margin at least $1/2$ and energy at most*

$$E_{\mathrm{pre}} + (t+13)\max\{c^2, 1\}^{1/(t+1)}.$$

Thus, for each fixed representable f, d, N, its strict-margin infimum energy satisfies $\kappa_{N,L}(f) = L + O_{f,d,N}(1)$ as $L \to \infty$.

Proof. Extend the scalar minimum $K_*(D, m)$ to any real $D > 0$: minimize $Da^2 + \sum_{j=2}^{m-1} b_j^2 + c'^2$ subject to $c' r_{m-1} \ge 1/2$, where $r_1 = \tanh a$ and $r_j = \tanh(b_j r_{j-1})$. The scalar baseline upper-bound argument works unchanged for real $D > 0$:

$$K_*(D, m) \le (m+12)(\max\{D, 4\}/4)^{1/m}.$$

Indeed, setting $q = (\max\{D,4\}/4)^{1/(2m)}$ and $r_j = q^j/[\sqrt{\max\{D,4\}}\min(j, m-j)]$ gives this bound by $a = \operatorname{atanh} r_1$, $b_j = \operatorname{atanh}(r_j)/r_{j-1}$, $c' = q$, and the summable reciprocal-ratio estimates used earlier.

Take the scalar tuple for $D = 4c^2$ and $m = t+1$. Replace the old readout by a hidden layer whose every row is $2av^T$, followed by the remaining scalar layers and readout. Its first new preactivation is $2az(x)$, whose absolute value is at least a and whose sign is $f(x)$. Monotonicity and oddness give the same sign with final margin at least $1/2$. The first new block has energy $4a^2c^2$; the remaining new blocks have exactly the remaining scalar costs. The total is therefore $E_{\mathrm{pre}} + K_*(4c^2, t+1)$, proving the displayed upper bound. An arbitrarily small readout increase supplies strict margins. Expanding this bound as $t \to \infty$, and combining it with $L(d/4)^{1/L} \le \kappa_{N,L}(f)$, proves the final assertion. □

These results leave the unrestricted Gaussian sample-complexity question open. They rule out exponential growth of minimum target energy under depth padding, while showing that bounded excess does not make every cheap realization robust. A target-relative argument must also control competing functions and their prior masses; subtracting a common transmission cost alone does not do so.

F Robust posterior compression uniformly for $\gamma \geq 1$

This section gives a different statistical bridge. It uses robustness of the original network while constructing a finite candidate, and tracks entropy only in the original weights that have not yet been processed. Consequently the intermediate compressed prefix need not have a density under a Gaussian prior. The result applies to a posterior containing many different functions.

Let $\mu_{N,t}$, $0 < t \leq 1$, have independent weight-entry variances t^2/d, t^2/N, and t^2/N^2 in the first, internal, and readout layers. The exponent-γ prior is obtained by $t = N^{(1-\gamma)/2}$. All-layer sign robustness refers to the total noise norm in (1); the clean $1/2$ margin may additionally be imposed.

Theorem F.1 (Uniform robust posterior compression)**.** *Fix a scalar 1-Lipschitz activation $|\phi| \leq 1$, and parameters $d \geq 1$, $L \geq 2$, $K \geq 1$, $0 < r \leq 1$, and $0 < \alpha \leq 1/8$. There is a finite class $\mathcal{H} = \mathcal{H}(d, L, K, r, \alpha)$, consisting of signs of bounded-weight depth-L networks, such that*

$$\log |\mathcal{H}| \leq \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

The polynomial is universal and does not depend on N or t. If $Q \ll \mu_{N,t}$ is supported on all-layer sign-r robust networks and

$$D_{\mathrm{KL}}(Q\|\mu_{N,t}) \leq NK/t^2,$$

then there is a distribution Γ_Q on $\mathcal{H}$ such that, for every $x \in \{-1,1\}^d$, a coupling of $w \sim Q$ and $h \sim \Gamma_Q$ satisfies

$$\mathbb{P}\{h(x) \neq \operatorname{sign} z_w(x)\} \leq \alpha.$$

The coupling may depend on x; its two marginals do not. Neither the support of Q nor its samples need represent a single Boolean function.

We give the proof, including deliberately loose polynomial choices. Two elementary entropy facts organize the construction. First, for a standard D-dimensional Gaussian G, the entropy variational inequality gives

$$\mathbb{E}_P\|Z\|^2 \leq 4D_{\mathrm{KL}}(P\|G) + 2D\log 2. \tag{7}$$

Second, Gaussian Talagrand [8] gives $W_2^2(P,G) \leq 2D_{\mathrm{KL}}(P\|G)$.

Lemma F.2 (Entropy of the unprocessed weights)**.** *Let A, X, Y be random variables, with fixed reference laws P_X, P_Y, and put*

$$B = \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y).$$

Let J be a finite message, possibly depending on (A, X, Y), and let $H = f(A, J, X)$. If A' is a randomized function of (A, J, H) whose fresh randomness is conditionally independent of Y, then

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'}\|P_Y) \leq B + H(J \mid A).$$

Proof. The left side equals $D_{\mathrm{KL}}(Q_Y\|P_Y)+I(Y;A')$. Data processing and

$$I(Y;J,H\mid A)\le I(Y;X,J\mid A)\le I(Y;X\mid A)+H(J\mid A)$$

bound it by $\mathbb{E}_{A,X}D_{\mathrm{KL}}(Q_{Y|A,X}\|P_Y)+H(J\mid A)$, which is at most the claimed bound by the KL chain rule. These expressions can equivalently be justified by finite partitions and lower semicontinuity. □

Here A will be a constructed prefix, X the current original weight block, and Y all later original blocks. The lemma imposes no reference law on A.

Lemma F.3 (Whole-layer active directions)**.** *Condition on (A,Y) and let X be a standardized Gaussian-reference matrix, with conditional relative entropy $R(A,Y)$. For vectors $b_x(A)$ of norm at most one, one can select a list J of inputs and its span $U=\operatorname{span}\{b_x : x\in J\}$ with*

$$\dim U\le\frac{t^2R(A,Y)}{N\delta}$$

such that preserving $H=tXU$ and replacing the orthogonal coordinates by fresh Gaussian coordinates changes the physical evaluation vectors tXb_x, under a suitable coupling, by expected squared RMS at most 2δ at each x. The candidate matrix law preserves (A,Y,H) and does not depend on x.

Proof. Use a fixed ordering of the finite input set. Whenever the averaged conditional KL of the entire residual evaluation vector Xv exceeds $N\delta/t^2$, add the unit residual direction v of that input. Each query is selected from the conditional law given (A,Y), averaging over active-coordinate values; thus J is a function of (A,Y). Orthogonal Gaussian coordinates are independent, so the active-coordinate entropy increases by precisely this conditional KL. The displayed rank bound follows. At termination, Gaussian Talagrand on the entire evaluation vector, multiplication by the squared residual-vector length, and division by N/t^2 give the error bound. Scalar-vector couplings extend to full matrices by disintegration. The basis of U is a measurable function of A,J under fixed Gram–Schmidt; it has no further dependence on Y once those variables are supplied. □

Parameter choices. Write

$$\begin{aligned}
e&=\alpha r^2/(1000L), & \eta&=\alpha/(100L), & q&=\lceil 2K/(e\eta)\rceil,\\
D_0&=[(d+1)q+2]\log 2, & V&=8K+2q, & B_H^2&=4V/e,\\
T&=128(q+1)(B_H^2+1)/e, & a&=e/[128(B_H^2+1)],\\
K_0&=(4K+2\log 2)/\eta, & B_v&=16K_0/r.
\end{aligned}$$

Choose m to be the least positive integer no smaller than both

$$\begin{aligned}
&10000T^4(q+1)^2a^{-2}[d+\log(10(q+1)/\eta)+1],\\
&10000B_v^2r^{-2}[d+\log(10/\eta)+1],
\end{aligned}$$

and put

$$N_0=\max\{m,\lceil LD_0/K\rceil,1\},\quad B_G=\max\{2,2\sqrt{\log(3m/e)}\},\quad B_1^2=4[4K+2(d+1)\log 2]/e.$$

All quantities have a fixed polynomial upper bound in the theorem parameters.

The small-width case. If $N \le N_0$, let $P_0 = N_0(d+1) + (L-2)N_0^2 + N_0$. Applying (7) to all standardized weights, and then multiplying by t^2, gives

$$\mathbb{E}_Q \|tZ_{\text{all}}\|^2 \le 4N_0K + 2P_0 \log 2.$$

With probability at least $1 - \alpha/4$, all actual weights are bounded by

$$B_{\text{small}} = [4(4N_0K + 2P_0 \log 2)/\alpha]^{1/2}.$$

Keep the original network on this event and use a default classifier otherwise. Padding with nodes having zero outgoing weights preserves the original output. The successful candidate has output margin greater than r. It remains to treat $N > N_0$.

The virtual prefix and its entropy budget. The constructed prefix A has N virtual nodes in its current layer. Each virtual row acts on m stored nodes of the preceding layer. The auxiliary description may keep all N rows; the final network will keep just the rows selected at the following stage. Track the conditional KL of the unprocessed *original* weights. By Lemma F.2, it is at most

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2. \tag{*}$$

This is not a KL bound for the singular compressed prefix itself.

For the first layer, zero every physical standardized row $tZ_{1,i}$ of norm greater than B_1, and retain all other rows. Bounded activations and (7) bound the expected squared RMS activation change at each input by e. This deterministic prefix channel does not increase the future entropy budget.

An internal layer: retain active data. Condition on the current prefix A and the future original weights Y. Apply Lemma F.3 to $b_x = h_A(x)/\sqrt{N}$ and $\delta = e$. Abort if more than q inputs are needed. By (*) and Markov, this failure has probability at most η. The successful query list, together with an abort flag, has a code of at most $[(d+1)q + 2]$ bits. Its entropy is at most D_0.

Let $H = tXU$ be the physical active matrix. Averaging the Gaussian entropy moment bound, with the success indicator rather than conditioning on success, gives

$$\mathbb{E}\left[\frac{\|H\|_F^2}{N};\ \text{success}\right] \le 4t^2 B_s/N + 2t^2 q \log 2 \le V.$$

Zero every row H_i whose norm exceeds B_H. The expected squared RMS activation change from this operation is at most $4V/B_H^2 = e$. All rows are clipped before any later selection, including selections that may depend on future weights.

An internal layer: reconstruct using m source nodes. For the orthonormal active basis let $u(j) = \sqrt{N}\, U_{j,:}$; uniform source sampling gives $\mathbb{E}_j u(j)u(j)^\top = I$. Clip the *whole vector* at radius T: $c(j) = u(j)\mathbf{1}_{\{\|u(j)\| \le T\}}$. Then

$$\mathbb{E}cc^\top \preceq I, \qquad \mathbb{E}\|u - c\| \le q/T.$$

Sample m uniform source indices S_j. Entrywise Hoeffding and a union bound give, with probability at least $1 - \eta$,

$$\|\widehat{G} - \mathbb{E}cc^\top\|_{\text{op}} \le a, \quad \sup_x \|\widehat{v}(x) - \mathbb{E}ch_A(x)\| \le a,$$
$$\sup_{x,y} |\widehat{K}(x,y) - K(x,y)| \le a,$$

where $\widehat{G} = m^{-1}\sum_j c(S_j)c(S_j)^\top$, $\widehat{v}(x) = m^{-1}\sum_j c(S_j)h_{A,S_j}(x)$, and $\widehat{K}$ is the empirical feature covariance; its population counterpart is $K(x,y) = N^{-1}\sum_i h_{A,i}(x)h_{A,i}(y)$. Abort on failure; this event depends only on A, J, S, so it does not reveal further future information.

Set $\widetilde{c}_j = c(S_j)/\sqrt{1+a}$, and let D have rows $\widetilde{c}_j^\top/\sqrt{m}$. Then $D^\top D \preceq I$ and $I - DD^\top \succeq 0$. Independently for every target virtual row draw $\xi_i \sim N(0, I - DD^\top)$ and use coefficients

$$\widehat{w}_{ij} = m^{-1}H_i^{\mathrm{clip}} \cdot \widetilde{c}_j + t\xi_{ij}/\sqrt{m}. \tag{8}$$

These are actual neural-network rows on the m sampled nodes. Their active feature vector is $\widehat{\psi}(x) = m^{-1}\sum_j \widetilde{c}_j h_{A,S_j}(x)$, and the lazy covariance is exactly $t^2(\widehat{K} - \widehat{\psi}^\top\widehat{\psi})$. For the ideal field, write $\psi(x) = U^\top b_x$ and $C(x,y) = K(x,y) - \psi(x)^\top\psi(y)$. Direct calculation gives

$$\sup_x \|\widehat{\psi}(x) - \psi(x)\| \le q/T + 2a =: \zeta, \qquad \sup_{x,y} |\widehat{C}(x,y) - C(x,y)| \le a + 2\zeta.$$

Here $\widehat{C} = \widehat{K} - \widehat{\psi}^\top\widehat{\psi}$. Bessel's inequality and $D^\top D \preceq I$ give $\|\psi(x)\|, \|\widehat{\psi}(x)\| \le 1$, which justify the covariance bound. At a fixed input, coupling scalar Gaussian evaluations with a common standard normal bounds the squared preactivation transport error, per row, by

$$B_H^2\zeta^2 + t^2(a + 2\zeta) \le e/8.$$

Extend these scalar couplings to full row fields. The candidate weight law is the same at every input; there is no claim of a single Gaussian-field coupling that is close at all inputs simultaneously.

Finally clip *all* coordinates ξ_{ij} to $[-B_G, B_G]$ before later row selection. Since each has variance at most one,

$$\mathbb{E}\left|\frac{t}{\sqrt{m}}\sum_j(\xi_{ij} - \mathrm{clip}_{B_G}\xi_{ij})h_{A,S_j}(x)\right|^2 \le 3me^{-B_G^2/4} \le e.$$

Thus every virtual coefficient is bounded by $B_H T/m + B_G/\sqrt{m}$, without a union bound over N.

All newly constructed data are a randomized function of A, J, H. Lemma F.2 therefore proves (*) for the next step. Combining entropy replacement, active clipping, finite reconstruction, and Gaussian clipping by $(a_1 + \cdots + a_4)^2 \le 4\sum_i a_i^2$, the expected squared RMS activation injection at this layer is less than $20e$.

The readout and final network. The original readout marginal has never changed. Its physical moment satisfies $\mathbb{E}_Q[N\|v\|^2] \le 4K + 2\log 2$. Abort if $N\|v\|^2 > K_0$, an event of probability at most η. Zero coefficients with $|Nv_i| > B_v$. This changes the output uniformly on bounded virtual feature vectors by at most $K_0/B_v = r/16$. Sampling m uniform last-layer sources and using readout coefficients $(Nv_i\mathbf{1}_{|Nv_i|\le B_v})/m$ adds at most $r/16$ uniformly on $\{-1,1\}^d$, except with probability η, by Hoeffding and the chosen m.

At each stage, retain in the final network just the m source rows selected from the preceding virtual layer. Each such row already acts on the previously stored m nodes. The result is an ordinary width-m network; repeated indices are duplicate nodes. Its weights are bounded by

$$B_{\mathrm{large}} = \max\{B_1, B_H T/m + B_G/\sqrt{m}, B_v/m, 1\}.$$

The larger virtual construction is not part of the final predictor.

The original network receives all perturbations. Fix an input x and glue the conditional couplings, retaining the original weight history. The current and future original blocks are untouched until their own stages. Define

$$\eta^s = h^s_{\text{virtual}}(x) - \phi(W^s_{\text{original}} h^{s-1}_{\text{virtual}}(x)).$$

The analogous readout difference is η^L. Later stages leave earlier computational prefixes unchanged, so the final candidate output is exactly the *original* network's output under these postactivation injections. No intermediate robustness assumption is made.

For precision, the canonical construction draws its new Gaussian coefficients independently of the discarded current block given (A, J, H). The comparison coupling may correlate them with that discarded block. It preserves the same joint marginal of the new prefix and the future original blocks, as well as the original full-network marginal, but need not preserve the canonical joint law with all discarded weights. The next entropy step uses the first marginal; original-network robustness uses the second. There is no conflicting demand for independence in the comparison coupling itself.

More explicitly, the induction invariant is that the comparison law has full original-weight marginal Q and the canonical marginal of (A, X, Y). The conditional transports supply a joint law of (A, X, Y, A') with this incoming marginal and the prescribed outgoing (A', Y) marginal. Extend it by the previous conditional law of the past original weights given (A, X, Y). The extension keeps the complete original weight law unchanged, preserves all earlier computational prefixes, and establishes the same invariant for the next layer. This is ordinary disintegration on finite-dimensional standard Borel spaces.

Abort probability is at most $2L\eta$. The unconditional moment bound gives

$$\mathbb{P}\left\{\text{no abort and } \sum_{s<L} \|\eta^s\|^2/N > r^2/8\right\} \le 160Le/r^2 = 0.16\alpha.$$

Otherwise the readout injection is at most $r/8$, and total squared noise is at most $9r^2/64 < r^2/4$. Robustness then gives candidate margin greater than $r/2$ relative to the original network's label: a smaller margin could be cancelled by an additional output shift within the radius-r noise ball. The exceptional probability is less than $\alpha/4$.

A fixed finite class. Pad candidates to width $M = N_0$ and take $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$. Round every parameter to a fixed grid on $[-B, B]$ of mesh at most

$$h = \frac{r}{16(d + L + 1)(2M(B + 1))^L}.$$

The bounded-activation, Lipschitz layer recursion gives uniform output change at most $r/8$. Thus rounding preserves every successful coupled prediction. Writing $P = M(d + 1) + (L - 2)M^2 + M$, the grid class satisfies

$$\log|\mathcal{H}| \le P \log\left[3 + \frac{32B(d + L + 1)(2M(B + 1))^L}{r}\right].$$

Although grid precision is exponentially small in L, its number of bits is polynomial in L. All other bounds above are polynomial too. Rounding is a fixed map on candidate networks, giving the common marginal Γ_Q and proving Theorem F.1.

Corollary F.4 (Polynomial-depth learning at every width)**.** *Suppose the raw Gaussian target event, consisting of networks with all-layer sign margin r and clean margin $1/2$ for f, has mass at least $\exp(-NK/t^2)$. Restrict the prior to all sign-r robust networks having global clean margin $1/2$, regardless of their Boolean function, then condition on n noiseless iid training examples. With probability at least $1-\beta$, the resulting posterior satisfies*

$$\mathrm{GibbsErr}(Q) \le 2\alpha + \sqrt{\frac{\log(2|\mathcal{H}|/\beta)}{2n}}.$$

Consequently the sample bound is jointly polynomial in d, L, K, r^{-1} and inverse desired error, uniformly in N and $0 < t \le 1$. For posterior-confidence error at threshold $2/3$, multiply Gibbs error by three; for threshold $1/2$, multiply it by two.

Proof. The posterior is the raw Gaussian conditioned on an event containing the target event, so its KL is at most NK/t^2. Its training error is zero. The compression couplings imply that Γ_Q has expected empirical error at most α. Finite-class uniform convergence holds for every member of $\mathcal{H}$, hence also for this sample-dependent mixture. Its true error is at most α plus the displayed square-root term. The pointwise coupling adds at most one more α to recover original posterior Gibbs error. This argument does not require a global coupling over inputs or a common function for all draws. The confidence-error statements follow from Markov on posterior error probability at a random test input. □

G A finite-radius certificate using clean Jacobian products

Products of individual spectral norms discard cancellations between layers. The following sufficient test retains the exact clean transition products and bounds the nonlinear remainder separately. It is a finite-radius statement, not an inference from the infinitesimal Jacobian alone.

Write the $H = L-1$ hidden layers in column-vector convention as $h_s = \tanh(W_s h_{s-1})$, with output $z = v^\top h_H$ and width N. Fix an input and its clean preactivations $a_s = W_s h_{s-1}$. Define

$$A_s = \mathrm{diag}(\mathrm{sech}^2 a_s) W_s, \qquad P_{s,t} = A_s \cdots A_{t+1} \ (t < s), \quad P_{s,s} = I.$$

The Gramian for normalized additive hidden disturbances is

$$G_1 = NI, \qquad G_s = A_s G_{s-1} A_s^\top + NI.$$

For a proposed total-noise radius r, recursively put $u_1 = 0$ and, for $s \ge 2$,

$$b_s = r\sqrt{\mathrm{diag}(W_s G_{s-1} W_s^\top)} + \sum_{t=2}^{s-1} |W_s P_{s-1,t}| u_t, \tag{9}$$

$$u_s = \frac{2}{3\sqrt{3}}\, b_s^{\odot 2}. \tag{10}$$

Absolute values, the square root, and the square in these expressions are coordinatewise. Finally define

$$B(r,x) = r\sqrt{v^\top G_H v + 1} + \sum_{t=2}^{H} |v^\top P_{H,t}| u_t. \tag{11}$$

Proposition G.1 (Finite-radius transport certificate)**.** *Every simultaneous postactivation and readout disturbance satisfying $\sum_{s=1}^{H} \|\eta_s\|^2/N + \eta_{\mathrm{out}}^2 \le r^2$ changes the output at x by at most $B(r,x)$. Consequently, if $B(r,x) < |z(x)|$ on the full input cube, the network preserves its own clean signs throughout this ball.*

Proof. Let Δh_s be the hidden displacement and write its exact recursion as

$$\Delta h_s = A_s \Delta h_{s-1} + \eta_s + R_s, \qquad \Delta h_s = \sum_{t=1}^{s} P_{s,t}\eta_t + \sum_{t=2}^{s} P_{s,t}R_t.$$

There is no input perturbation, so $R_1 = 0$. The global scalar bound $|\tanh''| \le 4/(3\sqrt{3})$ implies $|R_s| \le [2/(3\sqrt{3})]|W_s \Delta h_{s-1}|^{\odot 2}$. Assuming $|R_t| \le u_t$ at preceding layers, Cauchy–Schwarz applied to the linear disturbance map, followed by the coordinatewise triangle inequality for the remainder terms, gives $|W_s \Delta h_{s-1}| \le b_s$. This proves $|R_s| \le u_s$ inductively. At the readout, Cauchy–Schwarz including scalar output noise gives the first term of (11); the remainder sum gives the second. No statistical independence or small-noise approximation is used. □

The test can be sharpened by replacing the global curvature bound in (10) with the supremum of $|\tanh''|$ on $[a_{s,j} - b_{s,j}, a_{s,j} + b_{s,j}]$. An alternative exact scalar remainder envelope evaluates $\tanh(a+q) - \tanh(a) - \operatorname{sech}^2(a)q$ at $q = \pm b$ and at $q = -2a$ when $|2a| \le b$: these and $q = 0$ exhaust its extrema on the interval. The global-curvature version above already suffices for the numerical improvement below.

Application to the saved models. We evaluated the exact support condition $\min_x |z(x)| > 1/2$ and sign preservation at $r = .01/\sqrt{1 + S_N(w)}$ on all 36 saved width-32 models, using all 64 inputs of the six-bit cube. The preceding spectral/derivative-range test established membership for 28 models, excluded three by their clean absolute margin, and left five unresolved. The new test settles all five:

Task	Hidden layers, seed	Required r	$\max_x B(r,x)$	$\min_x \lvert z(x)\rvert$
Parity	$16, 2$	.000882	.012663	.996070
Majority	$64, 1$	.000943	.004762	.999996
Majority	$64, 2$	.000908	.006023	.999993
Parity	$32, 2$	.001018	.010793	.998686
Parity	$64, 0$	.000872	.007722	.999419

Thus all 33 successful fits pass a sufficient test for this precise support condition. The three remaining models are excluded because their minimum absolute clean outputs are .28125, .03125, and .06250, respectively. Their target misclassification alone was not used to infer failure of own-label robustness. The original cohort has 24/24 models passing; the separate depth-32/64 stress cohort has 9/12 passing and three excluded. No run was omitted, retrained, or subjected to new attacks for this check.

Numerical and empirical qualification. Two independently inspected implementations agree: one uses the Gramian and local curvature, while the other concatenates the exact clean disturbance transfer matrices and uses the global curvature bound. Clean linear gains also agree with independent backpropagation to relative discrepancy below 1.2×10^{-15}. All computations use float64, including spectral norms; they are numerical evaluations of rigorous sufficient tests, not interval-verified exact-real certificates. The earlier audit and all saved weights are retained. These finite examples demonstrate applicability to some gradient-trained networks through 64 hidden layers; they establish neither asymptotic polynomial depth scaling nor a universal property of SGD.

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